Venu Gopal Potnuru

Full Stack Developer | Assistant System Engineer
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SUMMARY

Full Stack Developer with 2+ years of experience building enterprise applications, REST APIs, platform integrations, and data-processing solutions using JavaScript, Node.js, Express.js, PostgreSQL, and Python. Experienced in performance optimization, API rate limiting, SIT/UAT, production deployment, and stakeholder collaboration. Built and deployed a local-first AI voice application using Rust, Tauri, Python, and speech recognition.

TECHNICAL SKILLS

Languages: JavaScript, Python, Rust, SQL, HTML, CSS
Frontend: React.js, JavaScript, Tailwind CSS
Backend: Node.js, Express.js, REST APIs, Python
Databases: PostgreSQL, SQLite
AI/Speech: NVIDIA Parakeet, Whisper, faster-whisper, Qwen TTS
Desktop: Rust, Tauri 2, CPAL
Tools: GitHub, Docker, Postman, AWS, Vercel, VS Code
Engineering: API Integration, Authentication, Rate Limiting, Caching, Performance Optimization, Data Processing

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) May 2024 – Present
Assistant System Engineer *Hyderabad, India*

Enterprise Learning Management Platform | Full Stack Developer

- Developed and maintained full-stack features for an enterprise learning platform integrated with **Cornerstone OnDemand (CSOD)**, supporting approximately **10K daily active users** and **200K peak load**.
- Built backend services using **Node.js and Express.js**, integrating CSOD APIs and developing custom internal APIs to support application-specific business requirements.
- Developed custom application screens, validations, business logic, workflow integrations, and **email notification triggers** using JavaScript, Tailwind CSS, and HTML.
- Reduced application/API load time from approximately **1 minute to consistently under 15 seconds** by identifying bottlenecks and optimizing application and API processing.
- Resolved API rate-limiting issues by implementing a custom rate-control mechanism using **jitter, progressive delays, and backoff strategies**.
- Designed and implemented data-processing solutions using **PostgreSQL and SQLite** for high-volume enterprise records and application workflows.
- Built automation to retrieve external data, process records, and load the resulting data into databases, reducing repetitive manual data-processing activities.
- Worked directly with clients and business stakeholders to understand requirements, clarify business rules, and deliver application changes.
- Participated in **SIT and UAT**, defect resolution, business validation, and independently handled application deployment activities.

PROJECTS

Voxa — Local-First AI Voice Typing Application [GitHub] *Rust, Tauri 2, Python, SQLite, NVIDIA Parakeet, Qwen TTS*

- Built a local-first Windows voice application that captures microphone audio, performs local speech recognition, and injects generated text into desktop applications.
- Implemented the speech-processing pipeline using **NVIDIA Parakeet** and built the application with **Rust, Tauri 2, and Python** for local AI inference.
- Developed the next version with voice activation, custom phrases, and custom meaning handling to improve recognition for Indian diction and usage patterns.
- Independently developed, packaged, deployed, and published downloadable releases for public testing.

Voxa Website — Product Showcase [Live Site] *React, Vite, Tailwind CSS, JavaScript, Vercel*

- Built and deployed a responsive React/Vite product showcase using **Tailwind CSS**, reusable components, and interactive product sections.
- Configured the production deployment using **Vercel** and maintained the public-facing website alongside the application releases.

EDUCATION

B.Tech in Computer Science and Engineering **2023 — 7.01 CGPA**
Avanthi Institute of Engineering and Technology